

Spring, 2026

MCT344-342: Industrial Robotics

Sheet 01: Review

Vectors:

- 1) A mobile robot is at $P_0 = (2, 1)$ m. It moves by displacement $d = (3, -0.5)$ m. **Find the new position P_1 .**
- 2) Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$.
 - a) Calculate: $A*B$ and $B*A$
 - b) Calculate: $B^{-1}.A^{-1}$
 - c) Calculate: $A^T.B^T$
- 3) Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ and $B = \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}$, calculate: $A*B$
- 4) If $u = (4, 0)$ and $v = (0, 3)$. Compute the dot product of the two vectors ($u \cdot v$) then decide if u and v are orthogonal/perpendicular.
- 5) A link points from $A = (1, 2)$ to $B = (5, 5)$. **Find the unit vector (direction) from A to B.**
- 6) If $A = (1, 3, 4)$ and $B = (2, 7, -5)$. Find the cross product ($A \times B$) then calculate the angle between A and B .

Matrices

- 7) Calculate the rotation matrix (R) to rotate the blue arrow to the green arrow by an angle theta as shown below in Fig.1.

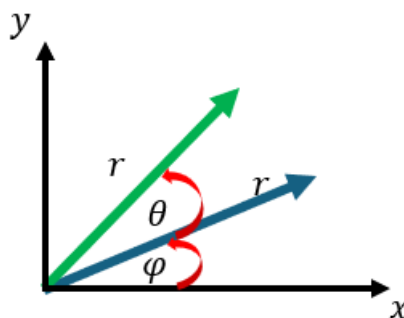


Fig.1

- 8) Find the total rotation matrix of a joint that first rotates by 45 degrees then by 30 degrees.

Trigonometry for planar linkages (sine rule + cosine rule)

- 9) A planar 2-link (2R) robot has link lengths $L_1 = 4$ cm and $L_2 = 3$ cm. The end-effector target point is at distance $d = 5$ cm horizontally from the base ($x = 0, y = 0$), target lies in ($x = 5, y = 0$). Find the elbow angle (the included angle between the two links).

- 10) A 2R robot has $L1 = 6$ cm, $L2 = 4$ cm. The target distance is $d = 11$ cm horizontally from the base ($x = 0, y = 0$), target lies in ($x = 11, y = 0$). Decide if the point is reachable and how many inverse kinematic solutions exist (0/1/2).
- 11) Consider the triangle in **Fig.2** where $a = 5$ cm, $b = 7$ cm, and angle $A = 30$ degrees. Use the sine rule to determine how many possible triangles exist, and find possible angles B .

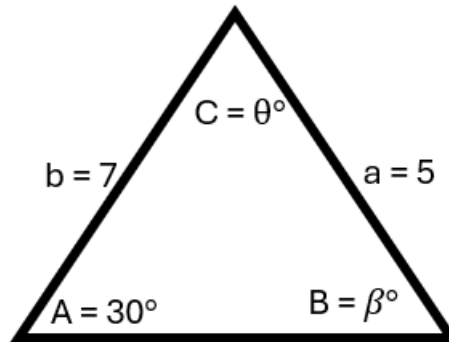


Fig.2

DOF Analysis

- 12) For each system, state the number of DOF:
- 2R planar arm (two revolute joints).
 - RP planar arm (one revolute + one prismatic).
 - Planar 4-bar linkage (four links including ground, four revolute joints).
- 13) Decide whether each is holonomic or nonholonomic:
- A robot end-effector must stay in a circle $x^2 + y^2 = 1$.
 - A differential-drive robot must satisfy sideways velocity = 0 (no lateral slip).
- 14) Regarding Redundancy and under-actuation:
- A 3R planar arm is used only to place its end-effector at (x, y). Is it redundant? If yes, by how many DOF?
 - A 2R arm has 2 DOF but only the first joint is actuated (second joint is passive). Is it underactuated?
- 15) Consider a planar four-bar linkage consisting of four rigid links connected in a single closed loop by four revolute (pin) joints.

Using the **Gruebler–Kutzbach** mobility criterion, determine the degrees of freedom (mobility) of the mechanism.

Assume:

- Motion is planar.
- All joints are 1-DOF revolute joints.

- 16) A Stewart–Gough platform shown in **Fig.3** consists of:

- A fixed base
- A moving platform
- 6 identical extensible legs connecting the base to the platform

Each leg has:

- 1 Prismatic joint (P) actuated (changes length)
- 2 Spherical joints (S) one at the base, one at the platform

So, each leg has a joint sequence:

$$S - P - S$$

Using mobility **Gruebler–Kutzbach** criterion for spatial mechanisms, determine:

What is the total number of degrees of freedom (DOF) of the moving platform relative to the base?

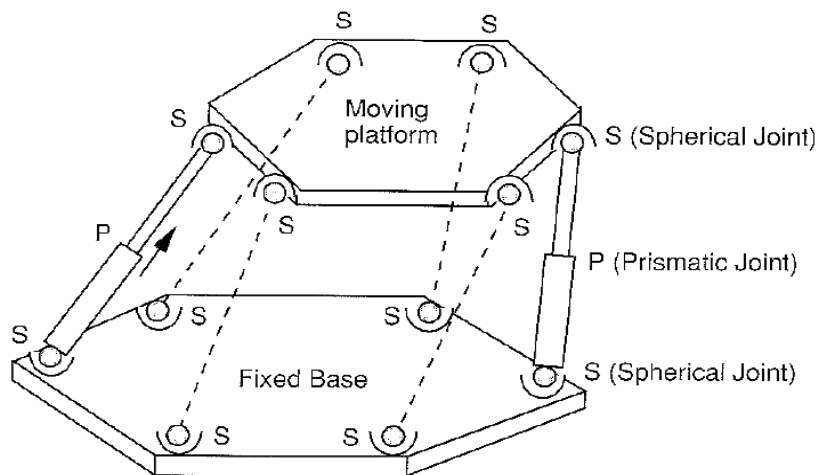


Fig.3 Stewart–Gough platform

graphical singularities in planar robotics

- 17) A 2R arm has $L1 = 80$ mm and $L2 = 60$ mm. Consider three targets P at distances from the base:
 - a) Case A: $d = 140$ mm
 - b) Case B: $d = 100$ mm
 - c) Case C: $d = 10$ mm
 For each case, determine graphically (using circles) the number of IK solutions and whether the configuration is singular.
- 18) Below are three configurations of a 2R arm. For each, decide if it is singular and explain why:
 - a) Arm fully stretched (links in a straight line).
 - b) Arm in a general bent posture.
 - c) Arm fully folded back (links in a straight line but opposite directions).
- 19) A four-bar linkage has ground pivots A and D. Link lengths are AB, BC, CD, and AD (ground). Graphical task: For a chosen position of B (input crank AB), construct point C as the intersection of circles:
 - Circle centered at B with radius BC
 - Circle centered at D with radius CD
 - a) Explain why there can be two assembly modes (two solutions) for C.
 - b) Show the singular “toggle” case on a drawing.

Forward & Inverse Kinematics

- 20) A planar 2R robot has link lengths $L_1 = 6$ cm and $L_2 = 4$ cm. Using a compass, draw its reachable workspace (no joint limits). Also compute the inner and outer radii.
- 21) Regarding the same robot ($L_1 = 6$ cm, $L_2 = 4$ cm) in the previous problem. Now joint-1 angle is limited to θ_1 in $[-30, 80]$ degrees. Draw the new workspace region (qualitatively) and explain the effect.
- 22) Ground pivots: A and D are fixed with $AD = 8$ cm on a horizontal line. Link lengths: $AB = 3$ cm, $BC = 7$ cm, $CD = 6$ cm. Input angle: AB makes 40 degrees above the ground line (from A).
Graphical task: locate B, then locate possible C by intersecting circle(B, BC) with circle(D, CD).
- 23) A 2R planar arm has $L_1 = 4$ cm, $L_2 = 3$ cm. Joint angles are $\theta_1 = 30$ degrees and $\theta_2 = 45$ degrees (θ_2 relative to link-1). Compute the end-effector position (x, y).
- 24) A 2R arm has $L_1 = 4$ cm, $L_2 = 3$ cm. Target is $P = (3, 4)$ cm.
 - a) Using compass: draw a circle centered at O with radius L_1 and a circle centered at P with radius L_2 .
 - b) How many intersections are there? What does that mean for IK solutions?
 - c) Use cosine rule to compute the elbow angle θ_2 .
- 25) Given in Fig.4 the lengths a, b, c, d and the angle θ_2 , find θ_3 and θ_4 .

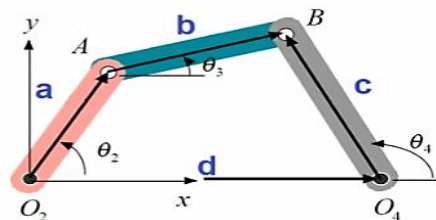


Fig.3 Four-bar mechanism

4-bar mechanism (motion feasibility test using Grashof's law).

- 26) A four-bar linkage has link lengths: 4 cm, 6 cm, 7 cm, and 9 cm. Determine whether the mechanism satisfies **Grashof's Law**. Identify the type of mechanism obtained.
- 27) A four-bar linkage has link lengths: 3 cm, 5 cm, 7 cm, and 10 cm. The 7 cm link is fixed. Determine whether the mechanism is crank-rocker, double-crank, or double-rocker.
- 28) A four-bar linkage has lengths 2 cm, 4 cm, 5 cm, and 6 cm. The shortest link (2 cm) is fixed. What type of motion occurs?
- 29) Using the same four-bar linkage that has lengths (2 cm, 4 cm, 5 cm, 6 cm), now fix the 6 cm link instead. What type of mechanism results?
- 30) A planar four-bar linkage has the following link lengths:
 - Link 1 (ground) = 6 cm
 - Link 2 = 4 cm

- Link 3 = 5 cm
- Link 4 = 7 cm
- a) Determine whether the mechanism satisfies **Grashof's Law**.
- b) Identify whether this is a **special (change-point) case**.
- c) Explain what happens to the mechanism when it reaches the singular configuration and why this is important in design.